Vector Spaces

What is a vector space?

n-dimensional Euclidean Space $\mathbb{R}^n$ is a vector space, as is $\mathbb{C}^n$.
$\mathbb{R}^n$ consists of vectors $\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ $x_i \in \mathbb{R}$ are called coordinates of $\underline{x}$.

We can add vectors in $\mathbb{R}^n$ and multiply them with scalars

Definition: A vector space V over $\mathbb{R}$ (or $\mathbb{C}$) is a set of elements $\underline{u}, \underline{v}, \underline{w}, \dots$ called vectors, together with:
- vector addition: given $\underline{u}, \underline{v} \in V$, there is another vector $(\underline{u}+\underline{v}) \in V$
- Scalar multiplication: given $\underline{u} \in V$ and $\lambda \in \mathbb{R}$ (or $\mathbb{C}$), there is another vector $\lambda\underline{u} \in V$

These operations satisfy:

(i) Commutativity: $\underline{u}+\underline{v} = \underline{v}+\underline{u}$
(ii) Associativity: $(\underline{u}+\underline{v})+\underline{w} = \underline{u}+(\underline{v}+\underline{w})$
(iii) There is a zero vector $\underline{0} \in \underline{V}$ satisfying $\underline{u}+\underline{0} = \underline{u}$ and there is, for every $\underline{u} \in V$, a vector $\underline{v} \in V$ s.t. $\underline{u}+\underline{v} = \underline{0}$
$\to V$ is a commutative group under addition $(\circ = +)$

Can be $\mathbb{C}$ for vector spaces over $\mathbb{C}$ instead of $\mathbb{R}$

(iv) Distributivity: $(\lambda+\mu)\underline{u} = \lambda\underline{u}+\mu\underline{u}$ for all $\lambda, \mu \in \mathbb{R}$ ($\mathbb{C}$) and all $\underline{u} \in V$
$\lambda(\underline{u}+\underline{v}) = \lambda\underline{u}+\lambda\underline{v}$ for all $\lambda \in \mathbb{R}$ ($\mathbb{C}$) and all $\underline{u}, \underline{v} \in V$
$\lambda(\mu\underline{u}) = (\lambda\mu)\underline{u}$ for all $\lambda, \mu \in \mathbb{R}$ ($\mathbb{C}$) and all $\underline{u} \in V$

(v) multiplication by $1 \in \mathbb{R}$ ($\mathbb{C}$) is identity: $1 \cdot \underline{u} = \underline{u}$ for all $\underline{u} \in V$

Examples of vector spaces

1. $\mathbb{R}^n$ and $\mathbb{C}^n$ – matrices with single column
2. All $m \times n$ real matrices $Mat_{m\times n}(\mathbb{R})$: $A, B \in Mat_{m\times n}(\mathbb{R}) \to (A+B) \in Mat_{m\times n}(\mathbb{R})$
$\lambda \in \mathbb{R}, A \in Mat_{m\times n}(\mathbb{R}) \to \lambda A \in Mat_{m\times n}(\mathbb{R})$
3. The set of all quadratic polynomials ax^2+bx+c, $a,b,c \in \mathbb{R}$:
We can add polynomials, and multiply them by scalars
4. The set of all functions $f: \mathbb{R} \mapsto \mathbb{R}$:
we can add: $(f+g)(x) = f(x)+g(x)$
we can multiply by $\lambda \in \mathbb{R}$: $(\lambda f)(x) = \lambda \cdot f(x)$
5. The set of vectors $\underline{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ satisfying $x+y+z=0$
↳ let $\underline{v}_1 = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$, $\underline{v}_2 = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} \to \underline{v}_1+\underline{v}_2 = \begin{pmatrix} x_1+x_2 \\ y_1+y_2 \\ z_1+z_2 \end{pmatrix}$
$\to x_1+y_1+z_1=0$ $\to x_2+y_2+z_2=0$
$(x_1+x_2)+(y_1+y_2)+(z_1+z_2) = \underbrace{(x_1+y_1+z_1)}_{=0}+\underbrace{(x_2+y_2+z_2)}_{=0} = 0$
↳ is in the vector space ☺
This is a subspace of $\mathbb{R}^n$.

Vector Subspaces

Definition: Let V be a vector space and let $\mathcal{U}$ be a non-empty subset of V.
Then $\mathcal{U}$ is called a vector subspace of V if $\mathcal{U}$ is itself a vector space (with the same addition and scalar multiplication)

Equivalently, a subset $\mathcal{U} \subseteq V$ is a vector subspace of V if:
- it contains the zero vector $\underline{0} \in \underline{\mathcal{U}}$
- if $\underline{u}, \underline{v} \in \mathcal{U}$, then $(\underline{u}+\underline{v}) \in \mathcal{U}$
- if $\underline{u} \in \mathcal{U}$ and $\lambda \in \mathbb{R}$ $(\mathbb{C})$, then $\lambda \underline{u} \in \mathcal{U}$

Examples of vector subspaces

Trivial examples:
- If V is a vector space, then $V \subseteq V$ is a vector subspace and $\{\underline{0} \in V\}$ is also a vector subspace of V.

More examples
- Set of $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$ satisfying $x+y+z=0$ is a subspace of $\mathbb{R}^3$
- The set of all solutions to a homogeneous system $A\underline{x} = \underline{0}$ is a vector subspace of $\mathbb{R}^n$ if $A \in Mat_{m\times n}(\mathbb{R})$
  - $\underset{Mat_{m\times n}(\mathbb{R})}{A} \times \underset{\mathbb{R}^n}{\underline{0}} = \underset{\mathbb{R}^m}{\underline{0}}$ $\rightarrow$ $\underline{0}$ is in this set of solutions
  - If $A\underline{x} = \underline{0}$ and $A\underline{y} = \underline{0}$, then $A(\underline{x}+\underline{y}) = A\underline{x} + A\underline{y} = \underline{0}+\underline{0} = \underline{0}$ — $x+y$ also a solution of $A\underline{x} = \underline{0}$
  - If $A\underline{x} = \underline{0}$, $\lambda \in \mathbb{R}$, then $A(\lambda \underline{x}) = \lambda \cdot A\underline{x} = \lambda \times \underline{0} = \underline{0}$ — $\lambda \underline{x}$ also a solution of $A\underline{x} = \underline{0}$
- $A \in Mat_{n\times m}(\mathbb{R})$, $\mathcal{U} = \{\underline{x} \in \mathbb{R}^m : A\underline{x} = \underline{0} \in \mathbb{R}^n\} \subset V = \mathbb{R}^m$
  - $\underline{0} \in \mathcal{U}$: $A\underline{0} = \underline{0}$ ✓
  - $\underline{u}, \underline{v} \in \mathcal{U}$ $(A\underline{u} = \underline{0}, A\underline{v} = \underline{0}) \Rightarrow A(\underline{u}+\underline{v}) = A\underline{u} + A\underline{v} = \underline{0}+\underline{0} = \underline{0} \Rightarrow (\underline{u}+\underline{v}) \in \mathcal{U}$ ✓
  - $\lambda \in \mathbb{R}$, $\underline{u} \in \mathcal{U}$ $(A\underline{u} = \underline{0}) \Rightarrow A(\lambda \underline{u}) = \lambda (A\underline{u}) = \lambda \times \underline{0} = \underline{0} \Rightarrow \lambda \underline{u} \in \mathcal{U}$ ✓
- $B \in Mat_{n\times m}(\mathbb{R})$. $\mathcal{U} = \{B\underline{z} \in \mathbb{R}^n : \underline{z} \in \mathbb{R}^m\} \subset \mathbb{R}^n$
  - $\underline{0} \in \mathcal{U}$: $\underline{z} = \underline{0} \in \mathbb{R}^m$: $B\underline{0} = \underline{0} \in \mathbb{R}^n$ ✓
  - $B\underline{z}_1, B\underline{z}_2 \in \mathcal{U}$: $B\underline{z}_1 + B\underline{z}_2 = B(\underline{z}_1+\underline{z}_2) \in \mathcal{U}$ ✓
    $\underline{z}_1, \underline{z}_2 \in \mathbb{R}^m$ $\quad$ $(\underline{z}_1+\underline{z}_2) \in \mathbb{R}^m$
  - $\lambda \in \mathbb{R}$, $B\underline{z} \in \mathcal{U} \Rightarrow \lambda B\underline{z} = B(\underset{\in \mathbb{R}^m}{\lambda \underline{z}}) \in \mathcal{U}$ ✓

Linear Independence, Spanning Sets and bases

Linear Independence

Let $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_n \in V$ be n vectors. A linear combination of these vectors is a sum

$\lambda_1 \underline{v}_1 + \lambda_2 \underline{v}_2 + \dots + \lambda_n \underline{v}_n \qquad \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$ (or $\mathbb{C}$)

Linear independence of $\underline{v}_1, \dots, \underline{v}_n$ means that none of these vectors can be written as a linear combination of the others.

Example: $\underline{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \underline{v}_2 = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$

Then $\underline{v}_3 = \begin{pmatrix} -1 \\ 0 \\ -3 \end{pmatrix}$ is not linearly independent of $\underline{v}_1, \underline{v}_2$, since $\underline{v}_3 = \underline{v}_1 - \underline{v}_2 = 1 \times \underline{v}_1 + (-1) \times \underline{v}_2$

Definition: Given a set of m vectors $S = \{\underline{v}_1, \underline{v}_2, \dots, \underline{v}_m\} \subset V$. We say S is linearly independent if $\lambda_1 \underline{v}_1 + \lambda_2 \underline{v}_2 + \dots + \lambda_m \underline{v}_m = \underline{0}$ implies $\lambda_1 = \lambda_2 = \dots = \lambda_m = 0$

(i.e. no linear combination exists to represent the zero vector except when $\lambda_1, \lambda_2, \dots, \lambda_m = 0$)

Assume $\underline{v}_1, \dots, \underline{v}_n$ are linearly dependent. Then there exists $\lambda_1, \dots, \lambda_m \in \mathbb{R}$, not all equal to zero, such that

$\lambda_1 \underline{v}_1 + \lambda_2 \underline{v}_2 + \dots + \lambda_m \underline{v}_m = \underline{0}$

Assume, without loss of generality, that $\lambda_1 \neq 0$. Then:

$\lambda_1 \underline{v}_1 = -\lambda_2 \underline{v}_2 - \lambda_3 \underline{v}_3 - \dots - \lambda_m \underline{v}_m$

$\underline{v}_1 = -\frac{\lambda_2}{\lambda_1} \underline{v}_2 - \frac{\lambda_3}{\lambda_1} \underline{v}_3 - \dots - \frac{\lambda_m}{\lambda_1} \underline{v}_m$

$\Rightarrow \underline{v}_1$ can be written as a linear combination of the other vectors

$\Rightarrow \underline{v}_1$ linearly dependent

Example: $S = \left\{ \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\} \subset \mathbb{R}^3$

$$\lambda_1 \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda_3 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$0\lambda_1 + 2\lambda_2 + \lambda_3 = 0$$
$$\lambda_1 + 0\lambda_2 + 2\lambda_3 = 0$$
$$2\lambda_1 + \lambda_2 + 0\lambda_3 = 0$$

Let $A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{pmatrix}$ $\rightarrow$ linear independence if unique solution

unique solution if $\det A \neq 0$

$\det A = -2 \begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = -2(-4) + 1 = 8 + 1 = 9 \neq 0$

$\Rightarrow A\begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{pmatrix} = \underline{0}$ has a unique solution

$\Rightarrow S$ linearly independent.

Example 2. $S = \left\{ \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}, \begin{pmatrix}1\\1\\1\end{pmatrix} \right\} \subset \mathbb{R}^3$

S is linearly dependent since $\begin{pmatrix}1\\0\\0\end{pmatrix} + \begin{pmatrix}0\\1\\0\end{pmatrix} + \begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}1\\1\\1\end{pmatrix}$

More generally, a set of more than n vectors in $\mathbb{R}^n$ is always linearly dependent

But: a set of less than n vectors in $\mathbb{R}^n$ is not always linearly independent, e.g. $\left\{ \begin{pmatrix}0\\0\\1\end{pmatrix}, \begin{pmatrix}0\\0\\2\end{pmatrix} \right\}$

A set containing the zero vector is always linearly dependent, since $\underline{0} = \lambda_1\underline{v}_1 + \lambda_2\underline{v}_2 + \dots + \lambda_n\underline{v}_n$ when $\lambda_1 = \lambda_2 = \lambda_n = 0$

A set of vectors is linearly independent if any vector $\underline{v} \in V$ can be written in at most one way as a linear combination of $\underline{v}_1, \dots, \underline{v}_n$

Spanning Set

A set $S = \{\underline{v}_1, \dots, \underline{v}_m\} \subset V$ is called spanning if every vector $\underline{v} \subset V$ can be expressed in at least one way as linear combinations of the vectors in S.

The span of a set S is the set of all linear combinations of the vectors in S

$$\operatorname{Span}(\{\underline{v}_1, \dots, \underline{v}_n\}) = \{\lambda_1\underline{v}_1 + \dots + \lambda_m\underline{v}_m : \lambda_1, \dots, \lambda_m \in \mathbb{R}\ (\mathbb{C})\}$$

S is spanning if $\operatorname{Span}(S) = V$.

Examples:

$S = \left\{ \begin{pmatrix}0\\1\\2\end{pmatrix}, \begin{pmatrix}2\\0\\1\end{pmatrix}, \begin{pmatrix}1\\2\\0\end{pmatrix} \right\}$

$\operatorname{Span}(S)$ is all $\begin{pmatrix}x\\y\\z\end{pmatrix} = \lambda_1\begin{pmatrix}0\\1\\2\end{pmatrix} + \lambda_2\begin{pmatrix}2\\0\\1\end{pmatrix} + \lambda_3\begin{pmatrix}1\\2\\0\end{pmatrix} = \begin{pmatrix}0&2&1\\1&0&2\\2&1&0\end{pmatrix}\begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{pmatrix}$

(for $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$)

S spans $\mathbb{R}^3$ if every vector $\begin{pmatrix}x\\y\\z\end{pmatrix}$ is a solution of $\begin{pmatrix}0&2&1\\1&0&2\\2&1&0\end{pmatrix}\begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{pmatrix} = \begin{pmatrix}x\\y\\z\end{pmatrix}$

If $\begin{pmatrix}0&2&1\\1&0&2\\2&1&0\end{pmatrix}$ invertible, then $\begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{pmatrix} = \begin{pmatrix}0&2&1\\1&0&2\\2&1&0\end{pmatrix}^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$, meaning for any $\begin{pmatrix}x\\y\\z\end{pmatrix}$ there exists a

$\begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{pmatrix}$ s.t. $\begin{pmatrix}x\\y\\z\end{pmatrix} = \lambda_1\begin{pmatrix}0\\1\\2\end{pmatrix} + \lambda_2\begin{pmatrix}2\\0\\1\end{pmatrix} + \lambda_3\begin{pmatrix}1\\2\\0\end{pmatrix}$

$\det \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{pmatrix} \neq 0$ so it is invertible

$\Rightarrow S$ spans $\mathbb{R}^3$

Example 2: $S = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \middle| \, x, y \in \mathbb{R} \right\} \subset \mathbb{R}^3$

S does not span $\mathbb{R}^3$

Fact: n vectors in $\mathbb{R}^m$ can never span $\mathbb{R}^m$ if $n < m$

A set of n vectors in $\mathbb{R}^m$ are always linearly dependent if $n > m$

Basis of a vector space

Linear independence $\approx>$ there are no superfluous vectors in the set

Spanning $\Rightarrow$ there are enough vectors to exhaust the vector space.

A basis set has both of these properties.

Definition: A set $S = \{\underline{v}_1, \dots, \underline{v}_n\} \subset V$ is a basis of V if S is both linearly independent and spanning.

Any basis of $\mathbb{R}^n$ consists of precisely n vectors because of

(This is true in any vector space, not just $\mathbb{R}^n$)

The cardinality of any basis of a vector space V is called the dimension, $\dim(V)$, of V.

Example: Standard basis of $\mathbb{R}^n$:

$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}$, $\dots$, $e_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ ← 1 is in the i^{th} row, $\dots$, $e_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$

Fact: n vectors $\underline{v}_1, \dots, \underline{v}_n$ in $\mathbb{R}^n$ form a basis iff $\det(A) \neq 0$, where $A = \begin{pmatrix} | & | & & | \\ \underline{v}_1 & \underline{v}_2 & \dots & \underline{v}_n \\ | & | & & | \end{pmatrix}$

i.e. the i^{th} column of $A = \underline{v}_i$

Example: The set of all functions $f: \mathbb{R} \mapsto \mathbb{R}$ forms a vector space V over $\mathbb{R}$

Let $U \subseteq V$ be the set of all polynomials of degree ≤ 3, e.g. $3x^2 - 5$

$\rightarrow U$ is a vector subspace of V

Any polynomial of degree at most 3 is of the form ax^3+bx^2+cx+d with $a,b,c,d \in \mathbb{R}$

A basis of U is $1, x, x^2, x^3 \Rightarrow \dim(U) = 4$

(V has infinite dimension so no finite basis exists)

Dimension and Coordinates, rank and unity

$A \in Mat_{m \times n}(\mathbb{R})$

$\{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{0}\}$ vector subspace of $\mathbb{R}^n$

$\{A\underline{z} \in \mathbb{R}^m : \underline{z} \in \mathbb{R}^n\}$ vector subspace of $\mathbb{R}^m$

Dimension

Theorem: Every vector space has a basis and the number of vectors in a basis for V is always the same, provided it is finite. This number is called the dimension of V, written $\dim(V)$.

Example:
$\dim(\mathbb{R}^n) = n$ with standard basis $\underline{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, \underline{e}_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$

Coordinates

Choose a basis $S = \{\underline{v}_1, \dots, \underline{v}_n\}$ for an n-dimensional vector space V. Then any vector $\underline{v}$ in V can be written in a unique way as $\underline{v} = c_1\underline{v}_1 + \dots + c_n\underline{v}_n$, with $c_1, \dots, c_n \in \mathbb{R}$ (or $\mathbb{C}$). We call $c_1, \dots, c_n$ the coordinates of V with respect to S.

We can think of a matrix $A \in Mat_{n \times m}(\mathbb{R})$ as a map $L_A : \mathbb{R}^m \to \mathbb{R}^n$
$\underline{x} \to A\underline{x}$

The domain of L_A is $\mathbb{R}^m$ and its image, denoted by $Im(L_A) = Im(A)$ is:

$$Im(A) := \{A\underline{x} \in \mathbb{R}^n \mid \underline{x} \in \mathbb{R}^m\} \subseteq \mathbb{R}^n$$

This is a vector subspace of $\mathbb{R}^n$

- $\underline{0} \in Im(A) \to \underline{0} = A\underline{0}$
- For $\underline{w}_1, \underline{w}_2 \in Im(A)$, need $\underline{w}_1 + \underline{w}_2 \in Im(A)$

$\underline{w}_1 = A\underline{x}_1$, $\underline{w}_2 = A\underline{x}_2$ for some $\underline{x}_1, \underline{x}_2 \in \mathbb{R}^m$
Then, $\underline{w}_1 + \underline{w}_2 = A(\underline{x}_1 + \underline{x}_2)$

- for $\lambda \in \mathbb{R}$ and $\underline{w} = Im(A)$, need $\lambda \underline{w} = Im(A)$
  $\underline{w} = A\underline{x}$ for some $\underline{x} \in \mathbb{R}^m$, then $\lambda \underline{w} = A(\lambda \underline{x})$

The dimension of $Im(A)$ is called the rank of A, denoted $Rk(A)$.

A subset of the domain of L_A is the kernel of L_A, denoted $Ker(L_A) = Ker(A)$, defined by:

$$Ker(A) = \{\underline{x} \in \mathbb{R}^m \mid A\underline{x} = \underline{0}\} \subseteq \mathbb{R}^m$$

This is a vector subspace of $\mathbb{R}^m$
It is the set of all solutions of the homogeneous equation $A\underline{x} = \underline{0}$.

The dimension of $Ker(A)$ is called the nullity of A.

Example: Compute kernel and nullity of $A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 3 & 2 & 1 & 7 \\ 4 & 3 & 4 & 7 \end{pmatrix}$

Consider $\left(\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 0 \\ 3 & 2 & 1 & 7 & 0 \\ 4 & 3 & 4 & 7 & 0 \end{array}\right)$ $\longrightarrow$ we are effectively solving $A\underline{x} = \underline{0}$ for $\underline{x}$, so Gaussian elimination

$$\Big\downarrow \begin{array}{l} R_2 - 3R_1 \\ R_3 - 4R_1 \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 0 \\ 0 & -4 & -2 & -2 & 0 \\ 0 & -5 & 0 & -5 & 0 \end{array}\right) \xrightarrow[-\frac{1}{5}R_3]{-\frac{1}{4}R_2} \left(\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array}\right) \xrightarrow{R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} & 0 \end{array}\right) \xrightarrow{-2R_3} \begin{array}{c} \left(\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right) \\ \begin{array}{cccc} x & y & z & w \end{array} \end{array}$$

$z - w = 0 \iff z = w$

Free variable $\lambda \rightarrow z = w = \lambda$
$y = -\lambda$
$x = -2\lambda$

$$Ker(A) = \left\{ \begin{pmatrix} -2\lambda \\ -\lambda \\ \lambda \\ \lambda \end{pmatrix} \in \mathbb{R}^4 \;\middle|\; \lambda \in \mathbb{R} \right\}$$

Note: Rank + nullity = no. of columns of matrix
nullity = no. of free variables.

Each $\underline{x} \in Ker(A)$ can be written as $\underline{x} = \lambda \begin{pmatrix} -2 \\ -1 \\ 1 \\ 1 \end{pmatrix}$

Nullity is 1 - kernel is a single basis vector and dim = no. of vectors in basis

Summary:

for $A \in Mat_{n\times m}(\mathbb{R})$

$Im(A) = \{ A\underline{x} \in \mathbb{R}^n \mid \underline{x} \in \mathbb{R}^m \}$

$rk(A) = dim(Im(A))$

$Ker(A) : \{ \underline{x} \in \mathbb{R}^m \mid A\underline{x} = 0 \}$

$null(A) = dim(ker(A))$

Examples:

- $A = 0$, the $n\times m$ zero matrix. That is, $A\underline{x} = 0$ holds for all $\underline{x} \in \mathbb{R}^m$
  $Im(A) = \{\underline{0}\} \subseteq \mathbb{R}^n$, $rk(A) = 0$, $ker(A) = \mathbb{R}^m$, $null(A) = m$

- $A = I_n$, the $n\times n$ identity matrix. $A\underline{x} = x \quad \forall x \in \mathbb{R}^n$
  $Im(A) = \mathbb{R}^n$, $rk(A) = n$, $Ker(A) = \{\underline{0}\}$, $null(A) = 0$

- $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in Mat_{2\times 3}(\mathbb{R})$

  $\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right)$

  Free var. λ

  $y = -2\lambda$

  $x = -2y - 3\lambda = \lambda$

  $Ker(A) = \{ \lambda \begin{pmatrix} 1 \\ -2 \end{pmatrix} \mid \lambda \in \mathbb{R} \}$

  $null(A) = 1$

- $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$

  $A\underline{v} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y + 3z \\ 2x + 4y + 6z \end{pmatrix} = (x + 2y + 3z)\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

  - $Im(A) = \{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} : \lambda \in \mathbb{R} \}$
    $= Span\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

  - $rk(A) = dim(Im(A)) = 1$

  - $ker(A)$ = all solutions of $x + 2y + 3z = 0$

    let $z = \lambda$, $y = \mu$

    $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \quad \lambda, \mu \in \mathbb{R}$

$$\ker(A) = \text{Span}\left\{\begin{pmatrix}-3\\0\\1\end{pmatrix}, \begin{pmatrix}-2\\1\\0\end{pmatrix}\right\}$$

$\Rightarrow \text{null}(A) = 2$

Rank - nullity Theorem: for $A \in \text{Mat}_{n\times m}$, $\text{rk}(A) + \text{null}(A) = m$ (m = number of columns)

Regarding image of $A = \begin{pmatrix} | & | & & | \\ \underline{v}_1 & \underline{v}_2 & \cdots & \underline{v}_m \\ | & | & & | \end{pmatrix}$, then $A\begin{pmatrix}x_1\\x_2\\\vdots\\x_m\end{pmatrix} = x_1\underline{v}_1 + x_2\underline{v}_2 + \dots + x_m\underline{v}_m$

$\Rightarrow \text{Im}(A) = \text{Span}\{\underline{v}_1, \dots, \underline{v}_m\}$ and $\text{rk}(A) = \dim(\text{Im}(A))$ is the number of linearly independent columns of A.

Proposition: $A \in \text{Mat}_{n\times m}(\mathbb{R})$

1. $\text{rk}(A)$ = number of linearly independent columns of A
2. $\text{rk}(A)$ = number of linearly independent rows of A.

$\Big\}\ \text{rk}(A) \le n,\ \text{rk}(A) \le m$

3. EROs don't change the rank, so $\text{rk}(A)$ = number of non-zero rows in RREF.

Examples:

$\text{rk}\begin{pmatrix}1&0\\0&1\end{pmatrix} = 2$ $\qquad \text{rk}\begin{pmatrix}1&2\\2&4\end{pmatrix} = 1$

$\text{rk}\begin{pmatrix}0&1&1\\1&0&1\end{pmatrix} = 2$ $\qquad \text{rk}\begin{pmatrix}0&1&1&3\\1&2&1&6\end{pmatrix} = 2$

$$A = \begin{pmatrix}1&2&3&4\\5&6&7&8\\9&10&11&12\end{pmatrix} \xrightarrow{\text{EROs}} \begin{pmatrix}1&0&-1&-2\\0&1&2&3\\0&0&0&0\end{pmatrix}$$

$\Rightarrow \text{rk}(A) = 2$

If $A \in \text{Mat}_n(\mathbb{R})$, A can only be reduced to the RREF I_n if A is invertible

If A invertible, then $\text{rk}(A) = n$, if A not invertible, then $\text{rk}(A) < n$

Fact: $\text{rk}(A)$ is the maximal size of an invertible square submatrix of A.

Example: $A = \begin{pmatrix}0&1&-2\\0&2&-4\\0&-3&6\end{pmatrix}$

$\det A = 0 \Rightarrow \text{rk}(A) \le 2$

all 2×2 submatrices have $\det = 0$

$\Rightarrow \text{rk}(A) \le 1$

Since A has non-zero entries $\Rightarrow \text{rk}(A) = 1$

Special Square matrices

$A=(a_{ij})$ is diagonal if $a_{ij}=0$ for all $i\neq j$ $\qquad \begin{pmatrix} a_{11} & & 0 \\ & a_{22} & \\ 0 & & \ddots & \\ & & & a_{nn} \end{pmatrix}$

- $\det(A) = a_{11}\times a_{22}\times\dots\times a_{nn}$
- If A and B diagonal then so is $AB \to AB$ has diagonal entries $a_{ii}b_{ii}$ and $AB=BA$.
- If A is diagonal with all $a_{ii}\neq 0$, then A^{-1} is a diagonal matrix with entries a_{ii}^{-1}

An $n\times n$ matrix $A=(a_{ij})$ is upper triangular if $a_{ij}=0$ when $i>j$ $\qquad \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ & 0 & & a_{nn} \end{pmatrix}$

- If A and B upper triangular, So is AB
- If A upper triangular and invertible, so is A^{-1}

(Same properties for lower triangular matrices.

1 2

3 4

Symmetric, Skew-Symmetric, Hermitian and anti-Hermitian matrices

A square matrix $A\in Mat_n(\mathbb{R})$ is Symmetric if $A=A^T$

A Square matrix $A\in Mat_n(\mathbb{R})$ is anti-symmetric if $A=-A^T$

Any matrix $A\in Mat_n(\mathbb{R})$ Can be written as $A=B+C$, with B symmetric and C anti-symmetric, where $B=\frac{1}{2}(A+A^T)$, $C=\frac{1}{2}(A-A^T)$

$A^{\dagger}$ = Complex Conjugate of the transpose of A.

$A=(a_{ij})$, $A^{\dagger}=(\overline{a_{ji}})$

A is Hermitian if $A^{\dagger}=A$

E.g. $A=\begin{pmatrix} 1 & 2+3i & -i \\ 2-3i & 5 & 3-i \\ i & 3+i & 5 \end{pmatrix}$, $A^{\dagger}=\begin{pmatrix} 1 & 2+3i & -i \\ 2-3i & 5 & 3-i \\ i & 3+i & 5 \end{pmatrix}=A$

$\therefore A$ is Hermitian

A is anti-Hermitian if $A^{\dagger}=-A$

E.g. $\begin{pmatrix} 3i & 2+3i & -i \\ -2+3i & 2i & 3-i \\ -i & -3-i & -5i \end{pmatrix}$

$A\in Mat_n(\mathbb{C})$ can be written as $A=\frac{1}{2}(A+A^{\dagger})+\frac{1}{2}(A-A^{\dagger})$

Orthogonal and Unitary Matrices

A matrix $A \in Mat_n(\mathbb{R})$ is called orthogonal if $A \times A^T = I_n$
A is orthogonal if $A^T = A^{-1}$
$\Big\}$ equivalent definitions

The set of all orthogonal $n \times n$ matrices is denoted $O(n)$. $O(n) \subset Mat_n(\mathbb{R})$ is a group with $\bullet$ = matrix multiplication.

If $A, B \in O(n)$, then $AB \in O(n)$ — Closedness

$A^T = A^{-1}$, $B^T = B^{-1}$

$(AB)^T = B^T A^T = B^{-1} A^{-1} = (AB)^{-1}$

$\therefore AB \in O(n)$ ✓

$I_n \in O(n)$ is the identity of $O(n)$

For every $A \in O(n)$, $A^T = A^{-1}$, so A is invertible with $A \times A^{-1} = A^{-1}A = I_n$, and $A^{-1} = A^T \in O(n)$

$$\underbrace{(A^T)^T}_{} = (A^{-1})^{-1} \Rightarrow \underset{\substack{\| \\ A^T}}{A^{-1}} \in O(n)$$

Examples of orthogonal matrices:

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Inner product:

$$\underset{\underline{v}}{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}} \cdot \underset{\underline{w}}{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}} = \sum_{i=1}^{n} x_i y_i = v^T w$$

$\underline{v} \cdot \underline{w} = 0 \iff \underline{v}$ and $\underline{w}$ orthogonal

$A = \begin{pmatrix} | & | & & | \\ \underline{v_1} & \underline{v_2} & \cdots & \underline{v_n} \\ | & | & & | \end{pmatrix} \in O(n)$ iff $\|v_i\| = 1$ and $\underline{v_i} \cdot \underline{v_j} = 0$ for $i \neq j$

$A = \begin{pmatrix} - \underline{w_1} - \\ - \underline{w_2} - \\ \vdots \\ - \underline{w_n} - \end{pmatrix} \in O(n)$ iff $\|\underline{w_i}\| = 1$ and $\underline{w_i} \cdot \underline{w_j} = 0$ for $i \neq j$

If $A \in O(n)$ and $\underline{v} \in \mathbb{R}^n$, then $||A\underline{v}|| = ||\underline{v}||$
and $\underline{v}_1, \underline{v}_2 \in \mathbb{R}^n$, then $(A\underline{v}_1)\cdot(A\underline{v}_2) = \underline{v}_1 \cdot \underline{v}_2$

$O(n)$ is called the orthogonal group.

A matrix $A \in Mat_n(\mathbb{C})$ is called unitary if $A \cdot A^+ = I_n$
A is unitary if $A^+ = A^{-1}$ $\quad \left(A^+ = (\overline{a_{ji}})\right)$

The set of all unitary matrices is denoted $\mathcal{U}(n)$ and is called the unitary group. $O(n)$ is a subgroup of this group.

Both orthogonal and unitary matrices belong to the set of normal matrices:
$A \in Mat_n(\mathbb{C})$ is called normal if $A \cdot A^+ = A^+ \cdot A$
Note: the set of normal matrices is not a group. -> Zero matrix is in this group but does not have an inverse.

<u>Eigenvalues, eigenvectors and eigenspaces</u>

$n \times n$ matrices map lines through the origin in $\mathbb{R}^n$ to lines through the origin, or to the origin itself.
So a 2×2 matrix could stretch a vector or reflect along a line -> eigenvectors with real eigenvalues
Or rotate around origin -> complex eigenvectors

In $\mathbb{R}^2$:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \quad A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

$A\begin{pmatrix} \lambda \\ 0 \end{pmatrix} = \begin{pmatrix} 2\lambda \\ 0 \end{pmatrix}$ -> Stretch by a factor of 2 in the x direction

$A\begin{pmatrix} 0 \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 4\lambda \end{pmatrix}$ -> Stretch by a factor of 4 in the y direction

Any vector $\begin{pmatrix} \lambda \\ 0 \end{pmatrix}$, $\lambda \neq 0$, is an eigenvector to the eigenvalue 2, any vector $\begin{pmatrix} 0 \\ \lambda \end{pmatrix}$, $\lambda \neq 0$, is an eigenvector to the eigenvalue 4.

Consider $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$

reflection in $y = x$

Any vector $\begin{pmatrix} x \\ x \end{pmatrix}$, $x \neq 0$ is mapped to itself : $A \begin{pmatrix} x \\ x \end{pmatrix} = \begin{pmatrix} x \\ x \end{pmatrix}$. Any such vector is an eigenvector of A to the eigenvalue 1.

Any vector $\begin{pmatrix} x \\ -x \end{pmatrix}$, $x \neq 0$, is mapped to $A \begin{pmatrix} x \\ -x \end{pmatrix} = \begin{pmatrix} -x \\ x \end{pmatrix} = -\begin{pmatrix} x \\ -x \end{pmatrix}$. Any such vector is an eigenvector of A to the eigenvalue -1.

Definition : Let A be a square matrix. A value λ is called an eigenvalue of A if there exists a non-zero vector $\underline{v}$ such that $A\underline{v} = \lambda \underline{v}$. Any such $\underline{v}$ is called an eigenvector of A to the eigenvalue λ.

If λ is an eigenvalue of A, then the eigenspace of λ is given by :

$V_\lambda = \{ \underline{v} \in \mathbb{R}^n \ (\mathbb{C}^n) : A\underline{v} = \lambda \underline{v} \}$

$V_\lambda \subset \mathbb{R}^n \ (\mathbb{C}^n)$ is a subspace of $\mathbb{R}^n$ and $\mathbb{C}^n$ and as such it contains $\underline{0}$.

-> $\underline{0} \in V_\lambda$

-> $\underline{v}, \underline{w} \in V_\lambda$: $A\underline{v} = \lambda \underline{v}$ and $A\underline{w} = \lambda \underline{w}$ $\Rightarrow A(\underline{v} + \underline{w}) = A\underline{v} + A\underline{w} = \lambda \underline{v} + \lambda \underline{w} = \lambda(\underline{v} + \underline{w})$

$\Rightarrow (\underline{v} + \underline{w}) \in V_\lambda$